Comparative Evaluation of ORB-Based BoVW, CNN, and Transfer Learning for Herbal Plant Recognition

CV Semester Project - Final Exam

Henry Ssenono 2025/HD05/26373/U
Paper ID *****

Jun 7, 2026

Abstract

- **Objective:** Automated recognition of herbal plants in diverse environmental conditions.
- **Dataset:** 425 images (14 classes) extracted from 21 video sequences.
- **Methods Investigated:**
  1. Traditional ML: ORB features + Bag-of-Visual-Words (BoVW).
  2. Deep Learning: Custom CNN trained from scratch.
  3. Transfer Learning: MobileNetV2 pre-trained on ImageNet.
- **Key Finding:** Transfer Learning achieved **93.23%** test accuracy, vastly outperforming handcrafted and shallow CNN methods.

Introduction

The Problem

Accurate herbal plant identification is vital for biodiversity, agriculture, and medicine, but remains difficult due to:

- Lighting variations and viewing angles.
- Complex backgrounds and scale changes.
- Intra-class variations (plants of same species looking different).

Our Contribution

A robust preprocessing pipeline and comparative study proving that pre-trained deep representations are superior for small botanical datasets.

- **Source:** 21 video recordings.
- **Classes:** 14 (Pseudo-labeled A–N).
- **Split:** 70% Train, 15% Val, 15% Test.
- **Preprocessing:** HSV-based vegetation filtering and Blur detection (Variance of Laplacian).

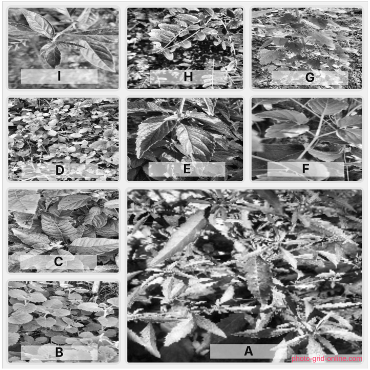


Figure 1: Sample herbal plant images from different classes.

Dataset Statistics

Table 1: Image count per pseudo-label class.

Class	**Count**	**Class**	**Count**
A	11	H	30
B	43	I	9
C	27	J	16
D	37	K	38
E	33	L	40
F	34	M	47
G	29	N	31
Total			**425**

Preprocessing Pipeline

- **Vegetation Filtering:** HSV color thresholding (green pixel ratio) to discard frames without plants.
- **Blur Detection:** Used the Laplacian operator to compute image variance; discarded low-frequency frames.
- **Normalization:** Resized to 224×224; scaled intensities to $[0, 1]$.
- **Data Augmentation:**
  - Horizontal flipping and $\pm 20^\circ$ rotation.
  - Brightness adjustments and Gaussian blur.

Approach 1: ORB + BoVW Pipeline

1. **Feature Extraction:** 1000 ORB keypoints per image (pool of 232,788 descriptors).
2. **Visual Vocabulary:** MiniBatch K-Means to create 500 "Visual Words."
3. **Classification:** Fixed-length histogram vectors fed into classical models.

Table 2: Traditional classifier validation results.

Model	Val Accuracy (%)
SVM	**47.92**
Logistic Regression	45.83
Random Forest	31.77
Gradient Boosting	28.65

Approach 2: Custom CNN

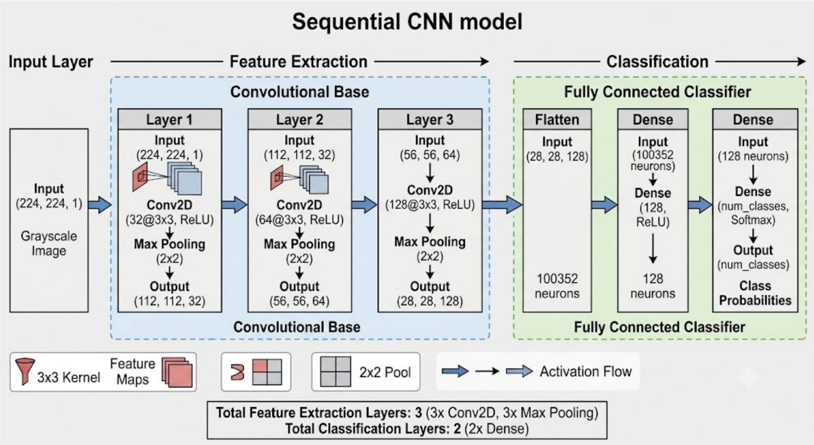


Figure 2: Architecture: 3 Conv Blocks (32, 64, 128) + 2 Dense Layers.

- Trained from scratch using Adam optimizer.
- Categorical cross-entropy loss.
- Softmax output for 14 classes.

Approach 3: MobileNetV2 Transfer Learning

- **Base:** MobileNetV2 pre-trained on ImageNet (Frozen).
- **Input:** Grayscale converted to 3-channel RGB.
- **Head:** GlobalAvgPooling $\rightarrow$ Dropout (0.5) $\rightarrow$ Dense (Softmax).

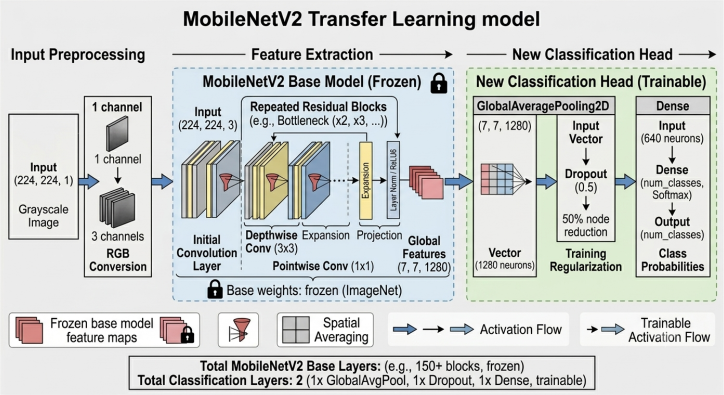


Figure 3: MobileNetV2 transfer learning architecture overview.

Final Model Comparison

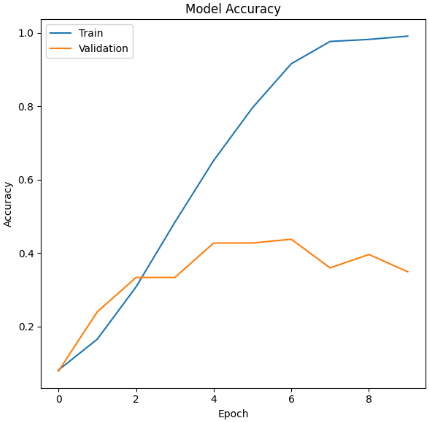
Table 3: Final comparison on the test set.

Method	Test Accuracy (%)
MobileNetV2 (Transfer Learning)	**93.23**
SVC (BoVW)	36.46
Custom CNN (From Scratch)	22.40

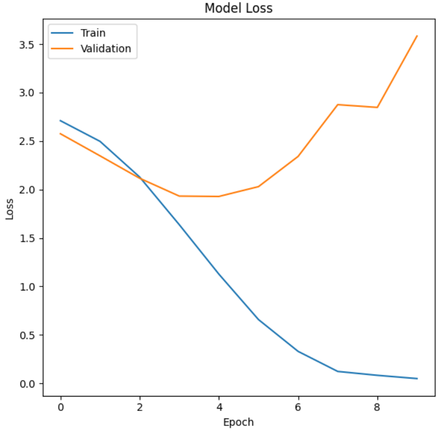
Observations

- Custom CNN suffered from overfitting due to small dataset size (425 images).
- Handcrafted ORB features failed to capture fine-grained botanical semantic details.

CNN Learning Curves



(a) Accuracy (Train/Val)



(b) Loss (Train/Val)

Figure 4: Custom CNN results showing significant overfitting/generalization gap.

Conclusion

- **Summary:** Transfer Learning is essential for botanical recognition when dataset size is limited.
- **Performance Gain:** MobileNetV2 improved accuracy by **+56.77%** over traditional BoVW.
- **Reproducibility:** The proposed preprocessing pipeline significantly improved image quality for feature learning.

Future Directions

- Expand dataset with more diverse environmental conditions.
- Investigate Vision Transformers (ViT) or EfficientNet.
- Real-time implementation for mobile plant identification.

Thank You!

CV Semester Project #*****